

Lakshmi Bharathy Kumar

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Authorized to work in the U.S. without sponsorship

SUMMARY

ML Engineer with an M.S. in Applied Data Intelligence from SJSU (May 2026) and 2 years of professional experience in data-driven marketing analytics. Focused on model training, evaluation, and deployment — from generative model fine-tuning and edge quantization through to cloud-ready inference pipelines on AWS.

TECHNICAL SKILLS

Languages: Python, SQL

ML & Deep Learning: PyTorch, Hugging Face, Transformers, scikit-learn, XGBoost, CNNs, LSTMs, GANs, Diffusion Models, LoRA/QLoRA, quantization, SHAP

Generative AI & Multimodal: Stable Diffusion, LangGraph, LangChain, RAG, CLIP, LLaVA, OpenCLIP, OpenAI / Gemini APIs

Cloud & MLOps: AWS (EC2, S3, Lambda, RDS, API Gateway, VPC, IAM, ECS, CloudWatch), Docker, FastAPI, ONNX, GitHub Actions (CI)

Data & Evaluation: Pandas, NumPy, ChromaDB, MongoDB, Snowflake, Streamlit, IS / CLIP / BLEU metrics

PROJECTS

Stable Diffusion Fine-Tuning: LoRA vs. QLoRA

PyTorch, Diffusers, Hugging Face, CLIP

- Fine-tuned SD v1.5 on 833 image-caption pairs; LoRA reached IS 1.18, CLIP 0.34 at 3.13 GB GPU / 11.5s while QLoRA matched scores at 4.10 GB / 16.2s — showing the memory benefit only materializes at larger model scales.
- Diagnosed underfitting at 7 effective optimizer steps across 36 generated images; compared IS and CLIP scores against human ratings (4.08/5) to trace where automated metrics diverge from perceived quality.

Multimodal Image Editing Agent

LangGraph, LLaVA 1.5-7B, InstructPix2Pix, OpenCLIP

- Built a 4-node LangGraph pipeline (preprocess, vision, prompt engineering, generation) using LLaVA 1.5-7B for grounded image understanding across analysis, style, enhancement, and ambiguous-prompt scenarios.
- Auto-evaluated outputs with OpenCLIP: text-image similarity 0.25–0.29, image-image similarity 0.91–0.95; 100% task completion across 12 runs with 5 triggering automatic revision loops.

Image Restoration Model for NPU Deployment

PyTorch, ONNX, Quantization, Edge AI

- Designed a lightweight residual CNN (1.2M params, 160 GFLOPs) achieving 21 dB PSNR at 256x256 on NPU hardware; exported to ONNX and compiled to MXQ via calibration-based quantization so the model runs entirely on-device with no API calls required.

GPT-Style Transformer from Scratch

PyTorch

- Built a 4-layer, 4-head character-level Transformer (174-token vocab, 128-dim, 512 FFN) on 100K TinyStories sequences; warmup and cosine decay training cut val loss from 1.50 to 0.96; benchmarked greedy vs. temperature sampling (T=0.8, 1.2) and documented 3 failure modes through systematic generation analysis.

Multimodal RAG System for PDF Q&A

ChromaDB, BM25, CLIP, MiniLM, Qwen 2.5

- Retrieves across text and images in a dense PDF using MiniLM + CLIP embeddings over 3 ChromaDB collections with BM25 and reciprocal-rank fusion; 4-bit quantized Qwen 2.5-1.5B generates grounded answers for all 15 benchmark questions on a single laptop.

PUBLICATIONS

L. B. Kumar, G. Liu, and Y. Luo, “Comparative Evaluation of Eye-Tracking Feature Representations for Gaze Direction Classification in Assistive Human–Machine Interaction,” *manuscript in preparation*, 2026.

EDUCATION

M.S., Applied Data Intelligence

May 2026

San Jose State University, San Jose, CA

- Thesis* — Engineered 5 eye-tracking feature types (raw gaze, fixation, UFF, heatmaps, eyeball images) from Pupil Core sensor data; benchmarked 5 ML classifiers and 2 fine-tuned CNNs (ResNet18, MobileNetV2) — RF hit 0.90 user-dependent, SVM/GBDT reached 0.94 cross-user (LOUO), 12 pp above the published UFF+RF baseline.

B.E., Electronics and Communication Engineering

2021

Saranathan College of Engineering, Trichy, India

EXPERIENCE

Digital Marketing Specialist — RepuNEXT, Chennai, India

Jun 2021 – Dec 2022

- Audited Google Analytics funnels to isolate drop-off points; updated page structure and SEO targeting based on findings, growing organic traffic 25%.
- Built automated reporting for weekly campaign metrics and audience segmentation; ROI improved 15%.